

Name-Hansika Kota

AI/ML Intern

Introduction

During the Raritone Summer Internship 2026, I worked on researching Artificial Intelligence applications in the fashion industry. The primary focus of my work was understanding how AI can improve online shopping experiences through personalized fashion recommendation systems and virtual try-on technologies.

Online fashion shopping has grown rapidly in recent years. However, customers often face challenges in selecting suitable clothing because they cannot physically try garments before purchasing. This frequently leads to dissatisfaction and product returns. AI-based recommendation systems and virtual try-on solutions aim to solve these problems by helping users discover products that match their preferences and visualize how garments may look on them.

The objective of this project was to study existing recommendation techniques, analyse state-of-the-art virtual try-on models, explore publicly available datasets, and understand how these technologies can be integrated into a complete fashion AI workflow.

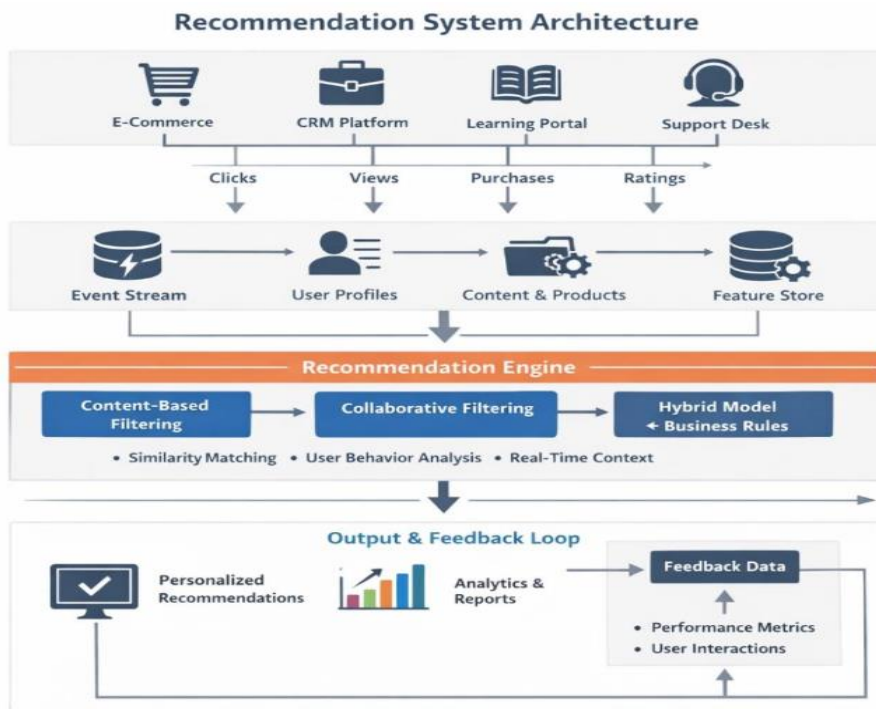
Research on Fashion Recommendation Systems

Fashion recommendation systems are designed to suggest products that are relevant to individual users based on their interests, browsing behavior, and purchase history.

The first technique studied was Content-Based Filtering. This method recommends products that are similar to items a user has previously viewed or liked. It focuses on product features such as colour, category, style, fabric type, and brand.

The second technique was Collaborative Filtering. Instead of focusing on product characteristics, this approach analyses user behaviour patterns. If two users have similar preferences, products liked by one user can be recommended to the other.

Modern fashion platforms increasingly use Deep Learning-based recommendation models. These models can learn complex patterns from large amounts of user and product data, leading to more personalized recommendations and improved customer satisfaction.



Datasets Explored

To understand how recommendation and virtual try-on systems are trained, I studied several publicly available fashion datasets.

DeepFashion is one of the most widely used datasets in fashion AI research. It contains thousands of clothing images with detailed annotations such as clothing categories, landmarks, and attributes.

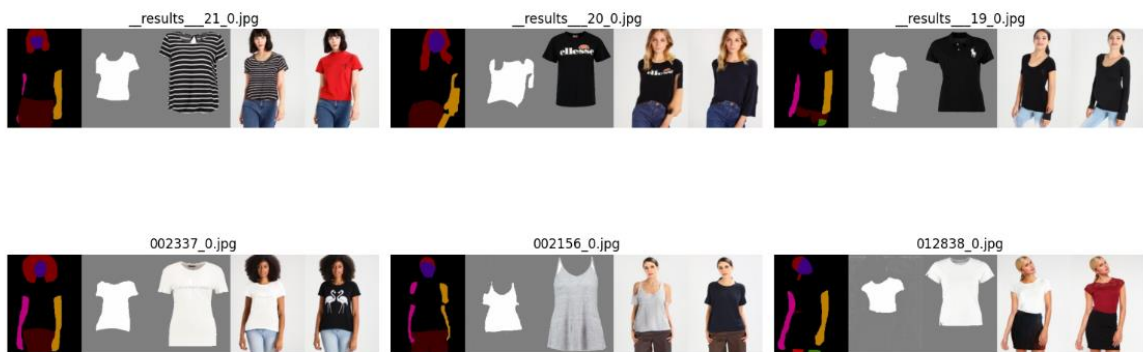
DeepFashion2 is an improved version that provides higher-quality annotations and more diverse clothing categories. It is commonly used for clothing detection, segmentation, and retrieval tasks.

The VITON dataset is specifically designed for virtual try-on research. It contains images of people and corresponding clothing items that can be used to train virtual try-on models.

Fashion-MNIST was also explored as a simple benchmark dataset for clothing image classification tasks.

These datasets play a crucial role in training and evaluating AI models for fashion applications.

DeepFashion Dataset Samples



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TESTING COMPLETE

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Research on Virtual Try-On Models

A major part of my internship involved studying virtual try-on technologies. Virtual try-on systems allow users to digitally visualize how a garment may appear when worn by a person.

The first model studied was VITON. It introduced the concept of warping clothing images to fit the body shape of a target person. While effective for its time, the generated images often lacked realism.

CP-VTON improved upon VITON by introducing better clothing alignment techniques. This helped preserve garment details and improved overall fitting quality.

HR-VITON focused on generating higher-resolution images and maintaining important clothing characteristics. The outputs were more realistic and visually appealing.

StableVITON represents a more recent advancement that utilizes diffusion-based image generation techniques. It produces highly realistic results while preserving garment structure and texture.

IDM-VTON is among the latest virtual try-on models and demonstrates impressive garment preservation, realistic fitting, and natural image generation. Based on my research, IDM-VTON and StableVITON currently provide some of the best virtual try-on results.

MODELS EXPLORED

VITON			CP-VTON			HR-VITON			StableVITON		
Input Person	Input Clothing	Output (Try-On)	Input Person	Input Clothing	Output (Try-On)	Input Person	Input Clothing	Output (Try-On)	Input Person	Input Clothing	Output (Try-On)
• One of the earliest virtual try-on models. • Transfers garments onto a person's image using pose information.			• Improves garment alignment and clothing warping. • Produces more natural fitting results.			• Focuses on preserving garment details. • Generates higher-quality and sharper outputs.			• Uses diffusion-based image generation. • Produces highly realistic and visually appealing results.		

2D and 3D Virtual Try-On Findings

I also compared 2D and 3D virtual try-on approaches.

2D virtual try-on systems work directly with images. They are computationally efficient and easier to implement. Most modern research models such as VITON, CP-VTON, HR-VITON, Stable VITON, and IDM-VTON belong to this category.

3D virtual try-on systems create a three-dimensional representation of the human body and garments. These systems provide more accurate fitting simulations and allow users to view clothing from multiple angles. However, they require greater computational resources, detailed body measurements, and more complex modeling techniques.

While 3D systems offer higher realism, 2D virtual try-on remains more practical for large-scale commercial applications due to lower computational costs.

2D VIRTUAL TRY-ON

2D virtual try-on generates the try-on result by directly transferring the garment onto a person's image.

How it works

Person Image



+

Garment Image



→

2D Try-On Result



Example Results



3D VIRTUAL TRY-ON

3D virtual try-on uses 3D models of the body and garments to generate realistic try-on results from multiple views.

How it works

3D Body Model



+

3D Garment Model



→

3D Simulation



→

Try-On Output
(Multi-view)



Example Results (Multi-view)



Proposed AI Workflow

Based on my research, I designed a workflow for an AI-powered fashion platform.

The process begins when a user uploads an image or enters preferences. User information is analysed to understand fashion interests and style preferences. A recommendation engine then suggests relevant products using hybrid recommendation techniques.

Once a garment is selected, body detection and segmentation models identify the person's body structure. The virtual try-on model then aligns and fits the selected garment onto the user image. Finally, the generated result is displayed, allowing the user to preview the outfit before making a purchase.

This workflow combines recommendation systems and virtual try-on technologies to create a more personalized shopping experience.

User Registration



User Preferences Collection

(Style, Color, Gender, Occasion)



Fashion Profile Creation



Recommendation Engine

(Content-Based + Collaborative Filtering)



Personalized Outfit Suggestions



User Feedback & Ratings



Recommendation Improvement



Better Future Recommendations

Challenges Faced

One of the major challenges was understanding the architectures of advanced virtual try-on models because many research papers contain complex deep learning concepts.

Another challenge was comparing multiple models developed using different datasets and evaluation metrics. This required careful analysis of research findings from various sources.

Managing large amounts of information and documentation was also challenging. To address this, I organized research papers, architecture diagrams, and comparison tables systematically.

Key Learnings

This internship helped me gain knowledge about recommendation systems, computer vision, deep learning, and generative AI applications in fashion technology.

I learned how recommendation algorithms personalize user experiences, how virtual try-on systems generate realistic clothing visualizations, and how modern AI models utilize large datasets to improve performance.

Additionally, I improved my research skills, technical documentation abilities, and understanding of current trends in AI-driven fashion solutions.

Future Improvements

Future fashion AI systems can incorporate real-time virtual try-on, improved body measurement estimation, personalized outfit generation, and AI-powered fashion assistants.

The integration of advanced diffusion models and large-scale recommendation systems can further improve realism and personalization. Combining recommendation engines with virtual try-on technologies will likely become a standard feature in future e-commerce platforms.

Conclusion

This internship provided valuable exposure to the practical applications of Artificial Intelligence in the fashion industry. Through research on recommendation systems, fashion datasets, and virtual try-on technologies, I developed a strong understanding of how AI can enhance online shopping and improve user experiences.

The study showed that modern virtual try-on models such as StableVITON and IDM-VTON are capable of generating highly realistic results, while recommendation systems help deliver personalized fashion suggestions based on user preferences.

Overall, this internship strengthened my knowledge of AI, deep learning, and computer vision while giving me practical insight into emerging technologies that are shaping the future of fashion and e-commerce.

Thank You, Team Raritone!

Thank you for providing me with the opportunity to learn, explore, and grow through this internship. The experience has been valuable in strengthening my understanding of AI, research, and emerging technologies in the fashion industry.

I am grateful for the support and guidance received throughout this journey.

– Hansika Kota
AI/ML Intern